1 A student wrote this comparison.

$$\boxed{?} > \frac{5}{2}$$

Which of these fractions belongs in the $\boxed{?}$ to make the student's comparison true?

Ⓐ $\frac{4}{2}$

Ⓑ $\frac{10}{4}$

Ⓒ $\frac{3}{5}$

Ⓓ $\frac{21}{6}$

2 Which of these numbers are prime numbers?

Select the **two** correct answers.

Ⓐ 4

Ⓑ 7

Ⓒ 18

Ⓓ 25

Ⓔ 43

This question has four parts. Be sure to label each part of your response.

3 A cafe owner recorded how many desserts were sold on Saturday.

- $\frac{5}{8}$ of the desserts sold were pies.
- $\frac{1}{8}$ of the desserts sold were cakes.
- The remaining desserts sold were cookies.

A. Which equation can be used to find the **total** fraction of desserts sold that were either pies or cakes?

Mark your answer by filling in the correct circle on the next page.

A $\frac{5}{8} + \frac{1}{8} = \frac{48}{16}$

B $\frac{5}{8} + \frac{1}{8} = \frac{6}{16}$

C $\frac{5}{8} + \frac{1}{8} = \frac{22}{8}$

D $\frac{5}{8} + \frac{1}{8} = \frac{6}{8}$

B. Of **all** the desserts sold on Saturday, what fraction were **cookies**? Show or explain how you got your answer.

C. Of all the desserts sold on Saturday, $\frac{4}{10}$ of the desserts were sold in the first two hours.

A worker at the cafe created this equation to represent the fraction of all the desserts sold in the first two hours.

$$\frac{3}{4} + \frac{1}{6} = \frac{4}{10}$$

Is the worker's equation correct? Show or explain how you got your answer.

D. At the beginning of the day on Saturday, the cafe had $6\frac{1}{8}$ pies to sell. At the end of the day, the cafe had $2\frac{3}{8}$ pies remaining.

What is the total amount of pie the cafe sold on Saturday? Show or explain how you got your answer.

Write your answers on the next page.

3 A. Ⓐ $\frac{5}{8} + \frac{1}{8} = \frac{48}{16}$

Ⓑ $\frac{5}{8} + \frac{1}{8} = \frac{6}{16}$

Ⓒ $\frac{5}{8} + \frac{1}{8} = \frac{22}{8}$

Ⓓ $\frac{5}{8} + \frac{1}{8} = \frac{6}{8}$

4 The floor of a room is in the shape of a rectangle. The floor has a length of 15 feet and a width of 12 feet.

What is the perimeter of the floor of the room?

Ⓐ 27 feet

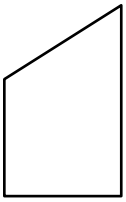
Ⓑ 27 square feet

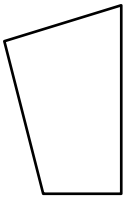
Ⓒ 54 feet

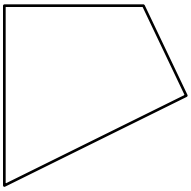
Ⓓ 54 square feet

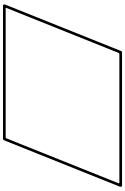
5 Which of these shapes appear to have **at least** one pair of parallel sides?

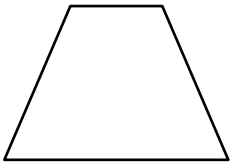
Select the **three** correct answers.

Ⓐ 

Ⓑ 

Ⓒ 

Ⓓ 

Ⓔ 

6 Which of these numbers is 231,198 rounded to the nearest **thousand**?

Ⓐ 230,000

Ⓑ 231,000

Ⓒ 232,000

Ⓓ 240,000

This question has two parts.

7 **Part A**

Which of these numbers in expanded form are shown with an equivalent number in word form?

Select the **three** correct answers.

Ⓐ (6 × 10,000) + (3 × 100) + (8 × 10) + (1 × 1)
sixty thousand, three hundred eighty-one

Ⓑ (6 × 10,000) + (3 × 100) + (8 × 10) + (1 × 1)
sixty-three thousand, eight hundred one

Ⓒ 60,000 + 3,000 + 800 + 1
six thousand, three hundred eighty-one

Ⓓ 60,000 + 3,000 + 800 + 1
sixty-three thousand, eight hundred one

Ⓔ (6 × 1,000) + (3 × 100) + (8 × 10) + (1 × 1)
six thousand, three hundred eighty-one

Ⓕ (6 × 1,000) + (3 × 100) + (8 × 10) + (1 × 1)
sixty thousand, three hundred eighty-one

Part B

Which of these statements are true?

Select the **three** correct answers.

Ⓐ Thirteen thousand, eight hundred thirty-one is **less than** 13,084.

Ⓑ Thirteen thousand, eight hundred thirty-one is **greater than** 13,084.

Ⓒ Thirteen thousand, seven is **less than** 13,084.

Ⓓ Thirteen thousand, seven is **greater than** 13,084.

Ⓔ Thirteen thousand, one hundred six is **less than** 13,084.

Ⓕ Thirteen thousand, one hundred six is **greater than** 13,084.

8 A teacher has 36 jars of paint to use in art class. The number of jars is 4 times the number of students in the class.

Which of these equations can be used to find y, the number of students in the class?

Select the **three** correct answers.

Ⓐ $36 \div 4 = y$

Ⓑ $4 \times y = 36$

Ⓒ $y \div 4 = 36$

Ⓓ $36 \times y = 4$

Ⓔ $36 \div y = 4$

9 What is the measure of an angle that turns through $\frac{1}{4}$ of a circle?

Ⓐ 15°

Ⓑ 90°

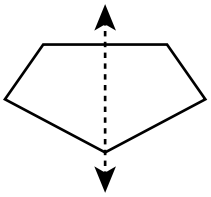
Ⓒ 270°

Ⓓ 345°

10 Which of these figures show a line that is a line of symmetry?

Select the **two** correct answers.

Ⓐ



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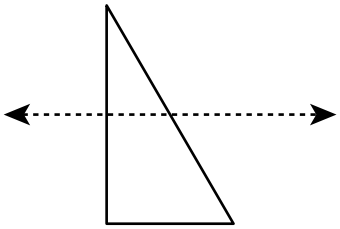

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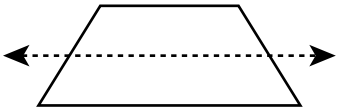

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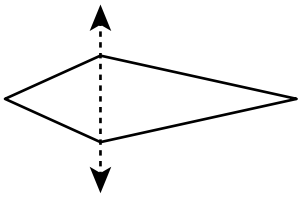

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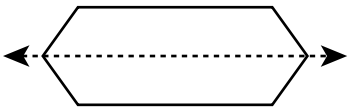


11 There are 5 jugs on a table. Each jug is filled with $\frac{3}{4}$ gallon of water.

What is the total number of gallons of water in all of the jugs?

Ⓐ $\frac{15}{4}$

Ⓑ $\frac{23}{4}$

Ⓒ $\frac{3}{20}$

Ⓓ $\frac{15}{20}$

12 An equation is shown.

$$70{,}368 - 2{,}419 = w$$

What value of w makes the equation true?

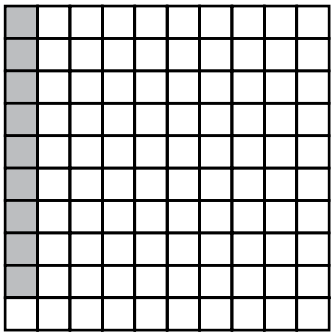
Ⓐ 67,849

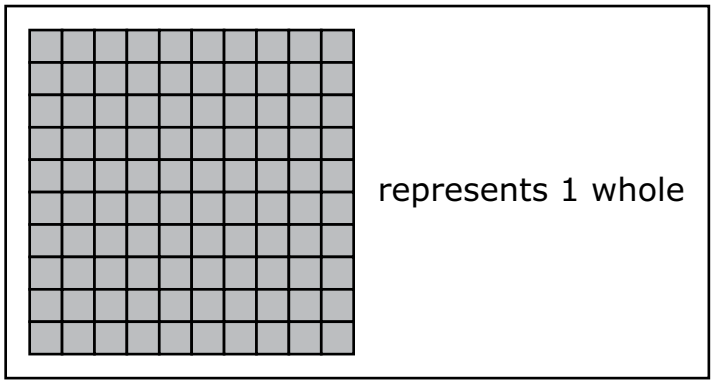
Ⓑ 67,949

Ⓒ 68,151

Ⓓ 68,949

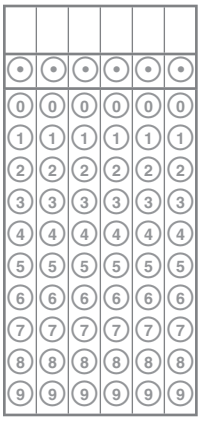
13 The shaded part of this model represents a decimal number that is less than 1.





Write a decimal number that is **less than** the decimal number represented in the model.

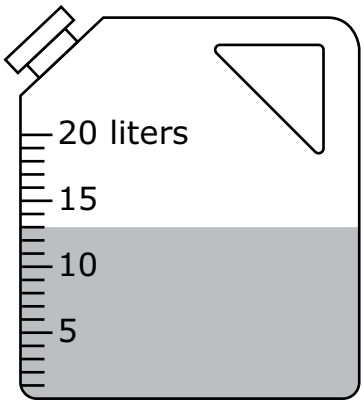
Enter your answer in the answer boxes at the top of the answer grid **and** completely fill the matching circles.



This question has four parts. Be sure to label each part of your response.

14 A gardener cut grass with a lawnmower and then raked and bagged leaves.

A. Before cutting the grass, the gardener added fuel to the lawnmower from a container. This picture shows the amount of fuel, in liters, **remaining** in the container after adding fuel to the lawnmower.



What is the total amount of fuel, in liters, remaining in the container?

B. The gardener used 2 liters of fuel to cut the grass.

What is the total amount of fuel, in **milliliters**, the gardener used to cut the grass? Show or explain how you got your answer.

C. It took the gardener 3 hours and 45 minutes to rake the leaves.

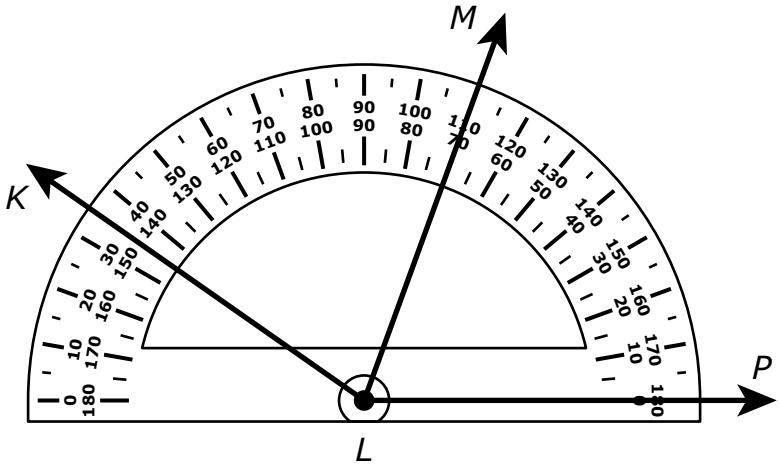
What is the total amount of time, in **minutes**, it took the gardener to rake the leaves? Show or explain how you got your answer.

D. The gardener placed the leaves into three large plastic bags: a black bag, a green bag, and a white bag.

- The mass of the black bag was 8 kilograms.
- The mass of the green bag was 7,900 grams.
- The mass of the white bag was **less than** the mass of the black bag but **more than** the mass of the green bag.

What could be the mass, in **grams**, of the **white** bag? Show or explain how you got your answer.

15 Angles *PLM* and *KLP* are shown on this protractor.



Which of these statements are true?

Select the **two** correct answers.

Ⓐ The measure of angle *PLM* is 35 degrees.

Ⓑ The measure of angle *PLM* is 70 degrees.

Ⓒ The measure of angle *PLM* is 110 degrees.

Ⓓ The measure of angle *KLP* is 45 degrees.

Ⓔ The measure of angle *KLP* is 145 degrees.

Ⓕ The measure of angle *KLP* is 155 degrees.

16 Which of these numbers are factors of 64?

Select the **three** correct answers.

Ⓐ 6

Ⓑ 8

Ⓒ 16

Ⓓ 24

Ⓔ 64

Ⓕ 128

17 A decimal number is shown.

0.49

Which fraction is equivalent to the decimal number?

Ⓐ $\frac{49}{1}$

Ⓑ $\frac{4}{9}$

Ⓒ $\frac{49}{10}$

Ⓓ $\frac{49}{100}$

18 What is the quotient of this expression?

4,527 ÷ 3

Ⓐ 1,509

Ⓑ 1,519

Ⓒ 1,566

Ⓓ 1,590

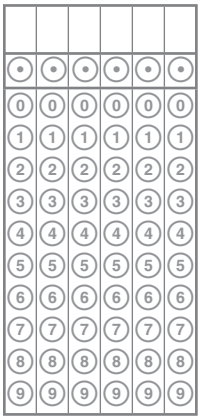
19 A student wrote this subtraction pattern.

234, 225, 216, 207, . . .

The student uses the same rule each time to find the next number in the pattern.

What is the next number in the student's pattern?

Enter your answer in the answer boxes at the top of the answer grid **and** completely fill the matching circles.



20 Which of these statements are correct?

Select the **three** correct answers.

Ⓐ 257 × 5 = 1,235 is true.

Ⓑ 257 × 5 = 1,235 is false.

Ⓒ 384 × 9 = 3,456 is true.

Ⓓ 384 × 9 = 3,456 is false.

Ⓔ 601 × 7 = 4,207 is true.

Ⓕ 601 × 7 = 4,207 is false.